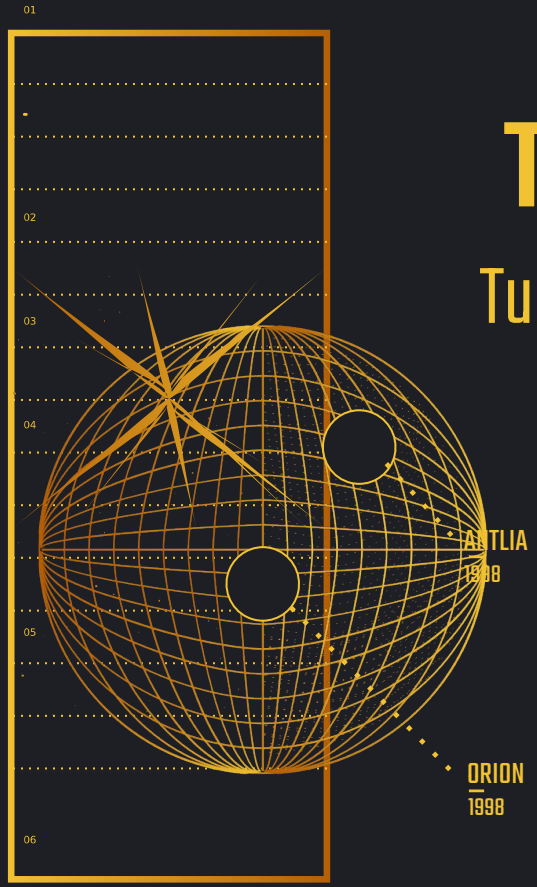

The Circumgalactic Medium

Turbulence, Magnetic Saturation, and Disk Instability

Presenters:
Param P. Gogia (Vassar),
Umman Azan (Middlebury), and
Sam Shelly (Vassar)

Advisor:
Edward Buie II

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Where are we, where are we going?

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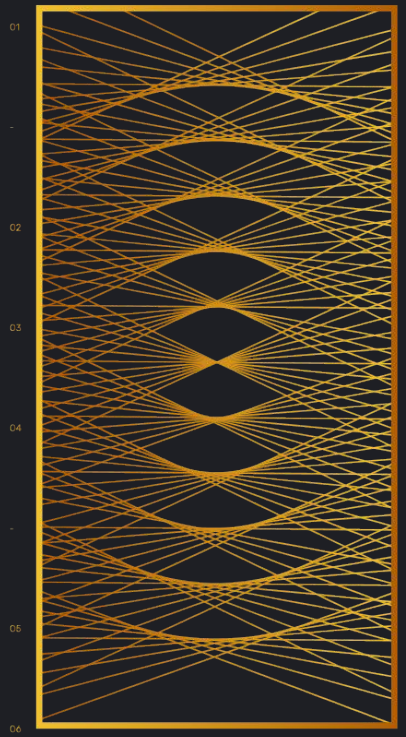
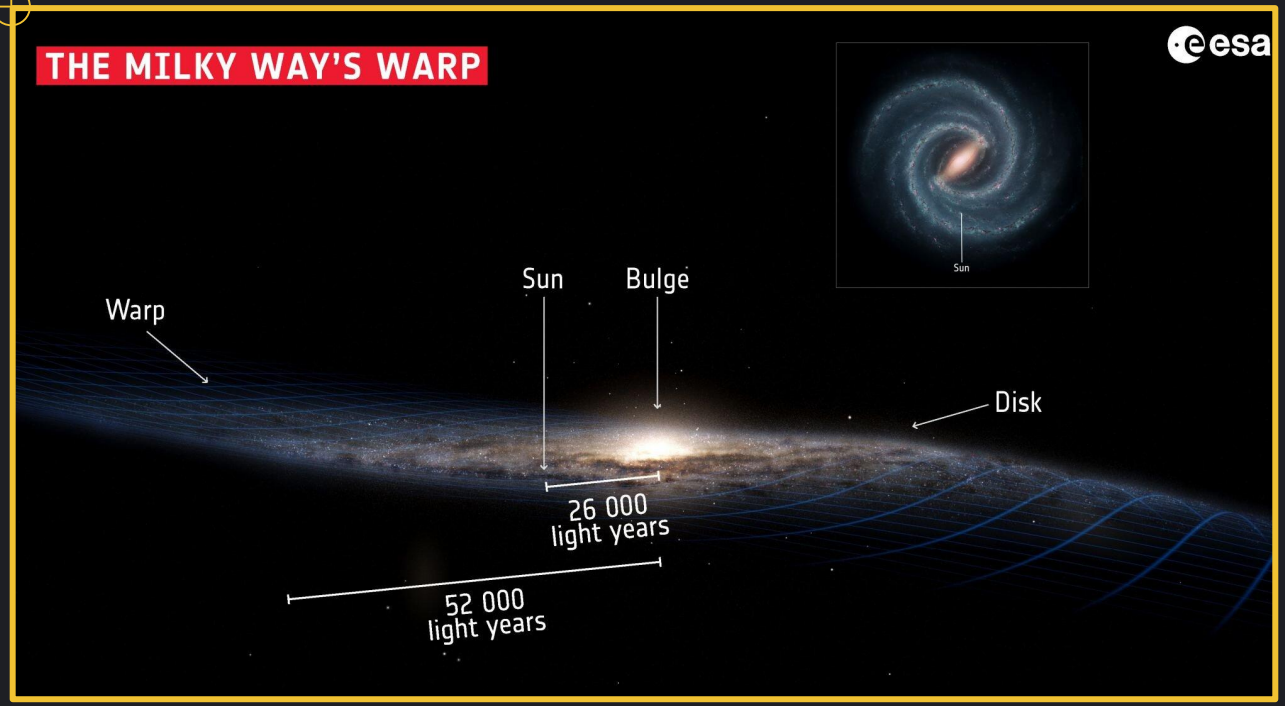


01.

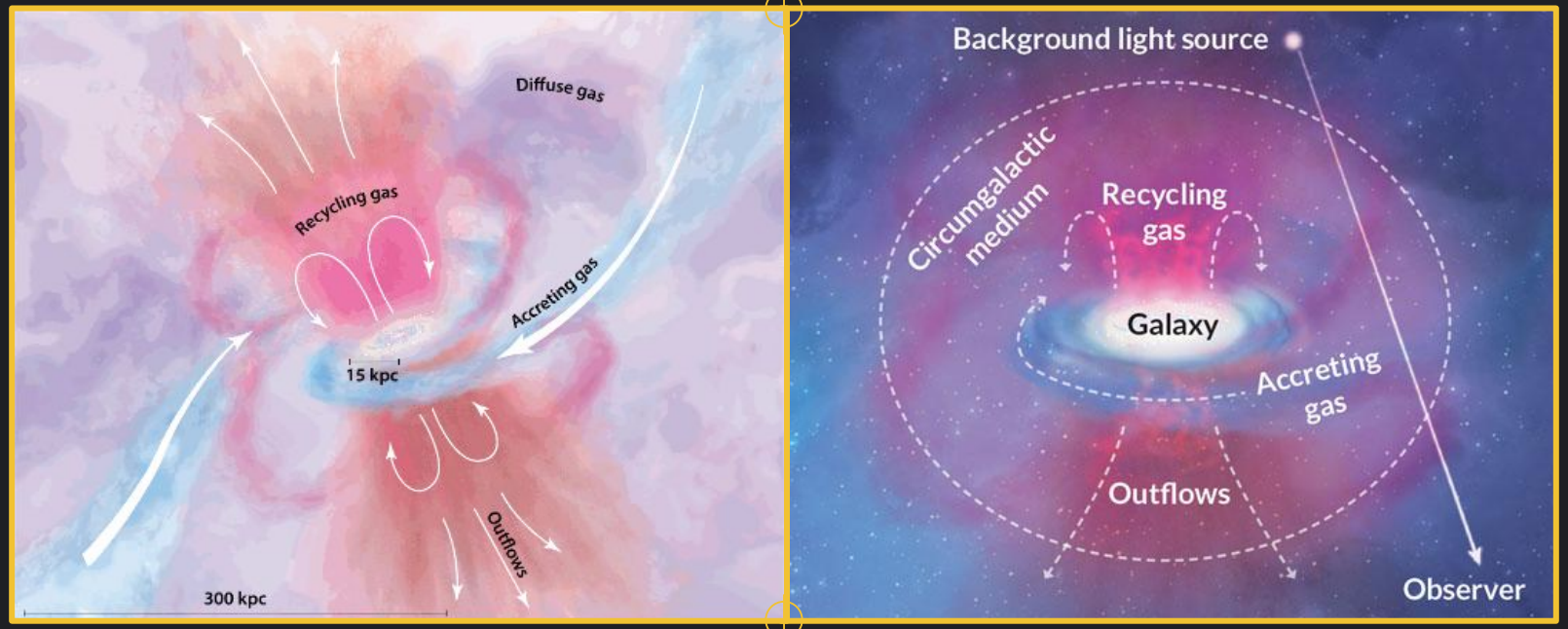
Background

The Circumgalactic Medium, its impact on galaxy formation and evolution

Key Features of Galaxies



The Circumgalactic Medium (CGM henceforth)

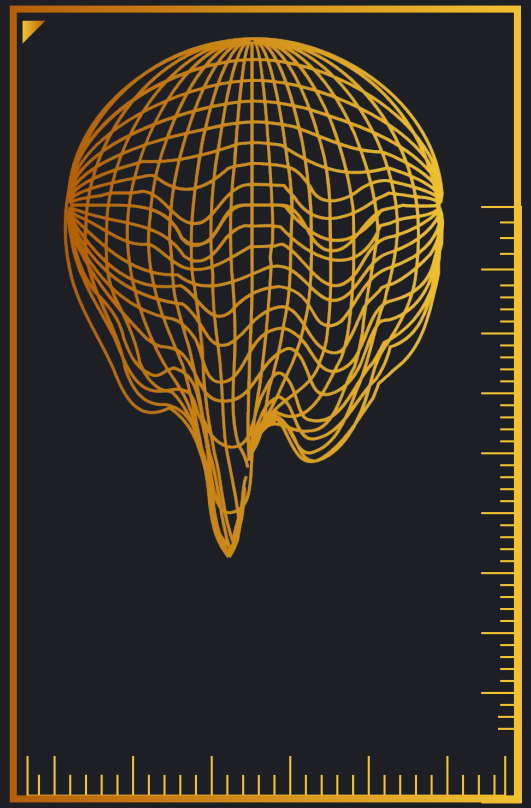


Tomlinson, et. al. (2017)

What does the CGM Tells us About Galaxies?

- Diffuse, multiphase gas outside the disk but inside the virial radius.
- Robust dynamical behaviour with starburst activity, supernovae, AGN feedback, and galactic accretion.
- Controls galaxy formation and evolution.
- Chemically rich: multiple ionic species, varying ionisation levels, rich distribution
- Magnetic fields observed to dynamically relevant strengths (order of microGauss)

Observations are hard! QSO spectral features are uncertain and limited in scope. Faraday Rotation maps are too specific.



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02.

Our Methods

The place of Simulations in the Scientific Method and MAIHEM

Simulations and Science

The diffuse nature of CGM makes it hard to observe, simulations are a natural alternative!

+ Freedom over initial conditions, the parameter space, fairly accurate, flexible Physics
+ Inexpensive: economically, temporally, spatially
+ Theory takes you as far as you want to go…

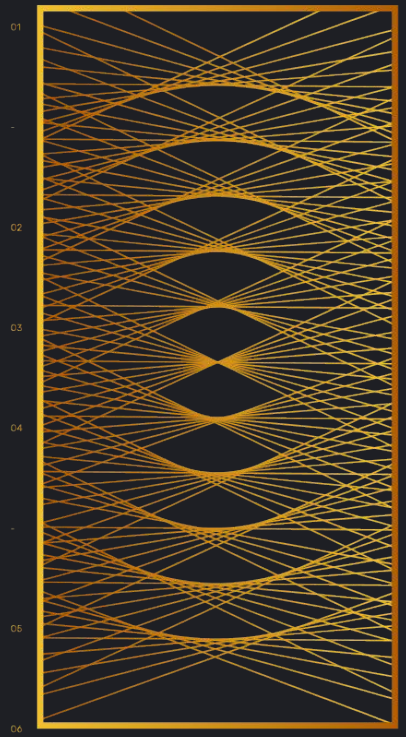
- Resolution
± New Physics!

Introduction to MAIHEM

Models of Agitated and Illuminated Hindering and Emitting Media:

1. Non-equilibrium chemistry
2. Robust Physics
3. Magneto-hydrodynamics
4. Excellent spatial and temporal resolution

Prof. Buie's modifications:
Consider the NFW CDM density profile
Inject power-law turbulence injection profile
Seed B field in the Z direction



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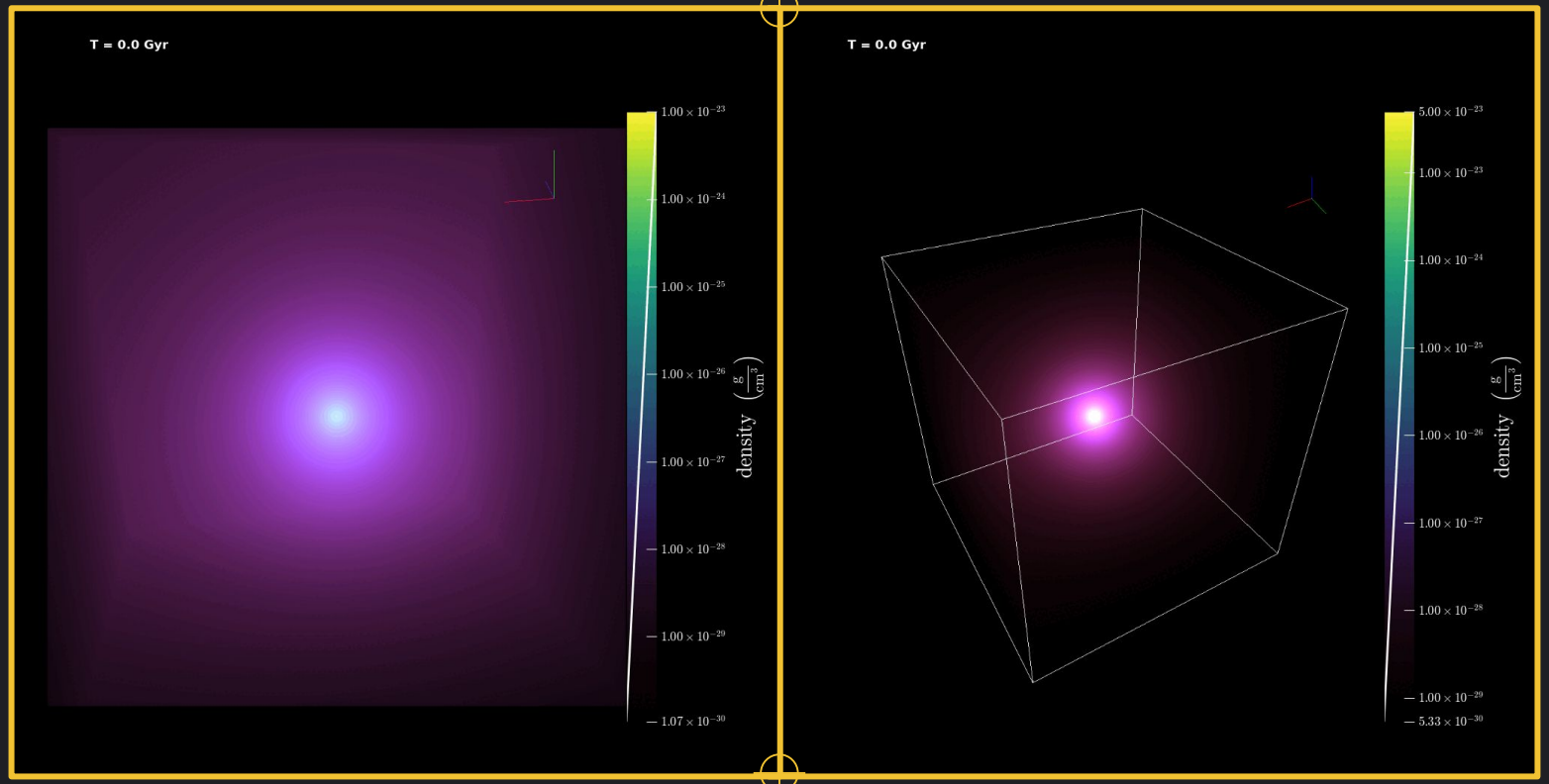
Results!

‘The pains of being a theorist…’

– Ed Buie II, circa 2024 BCE

Overview of the Simulations: The Impact of Turbulent Feedback



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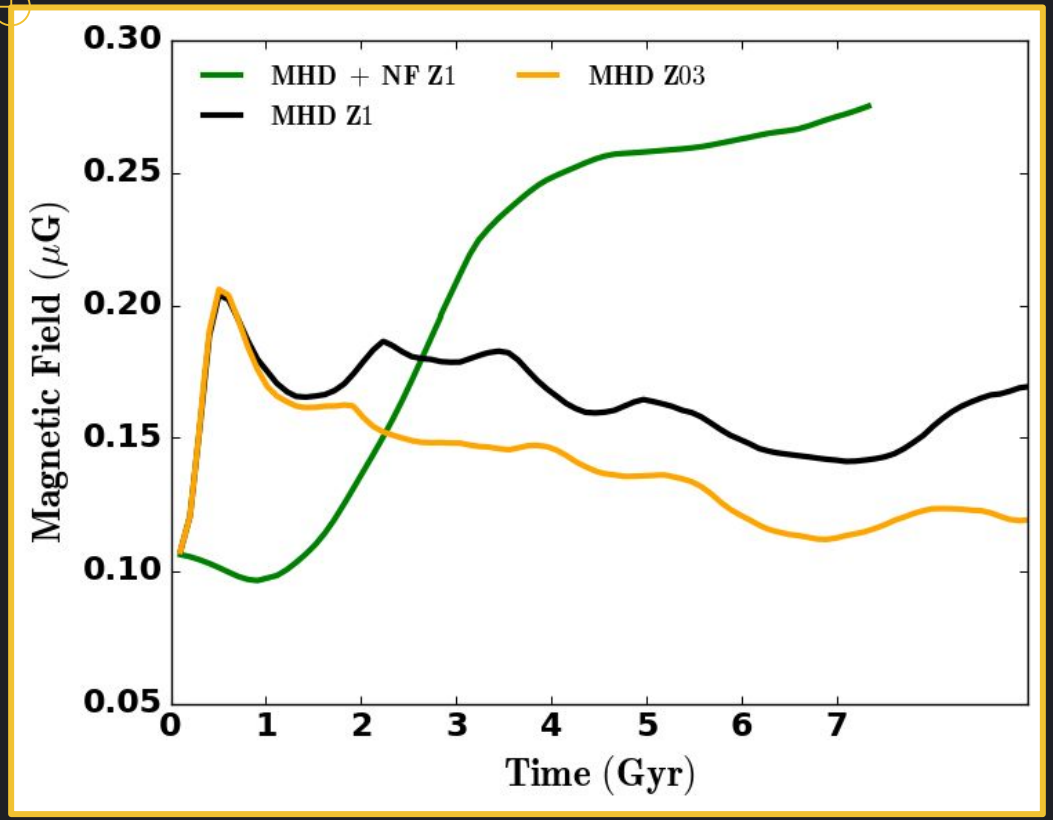
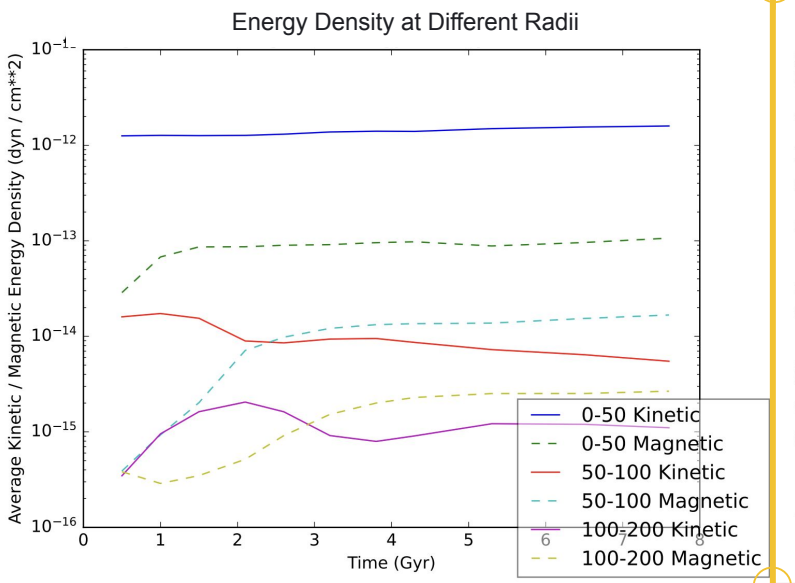
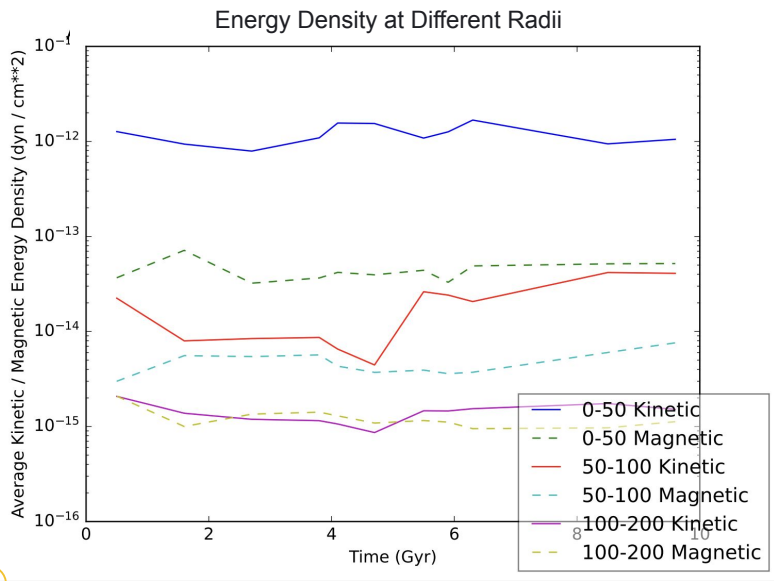
Turbulence Causes the Magnetic Field to Saturate(...?)

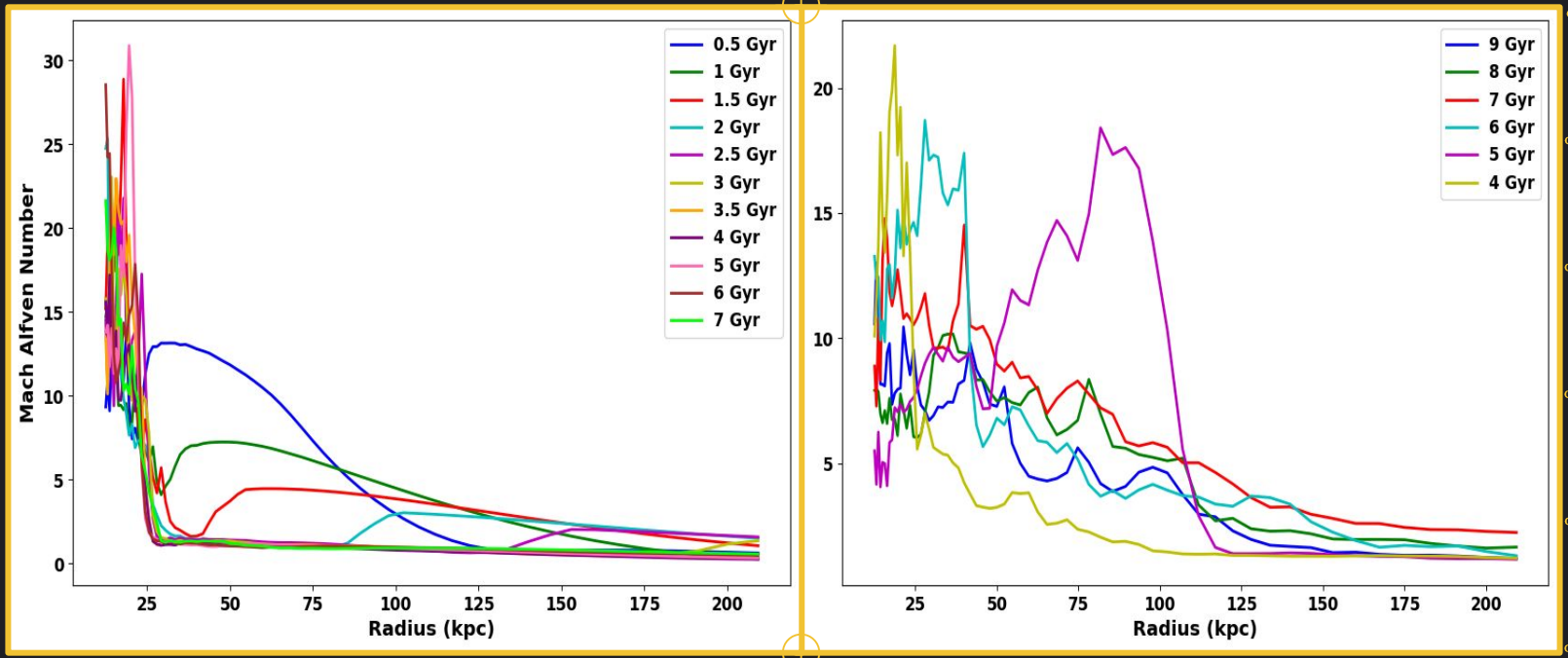


Which Radial Interval in the CGM Dominates this Saturation Process?

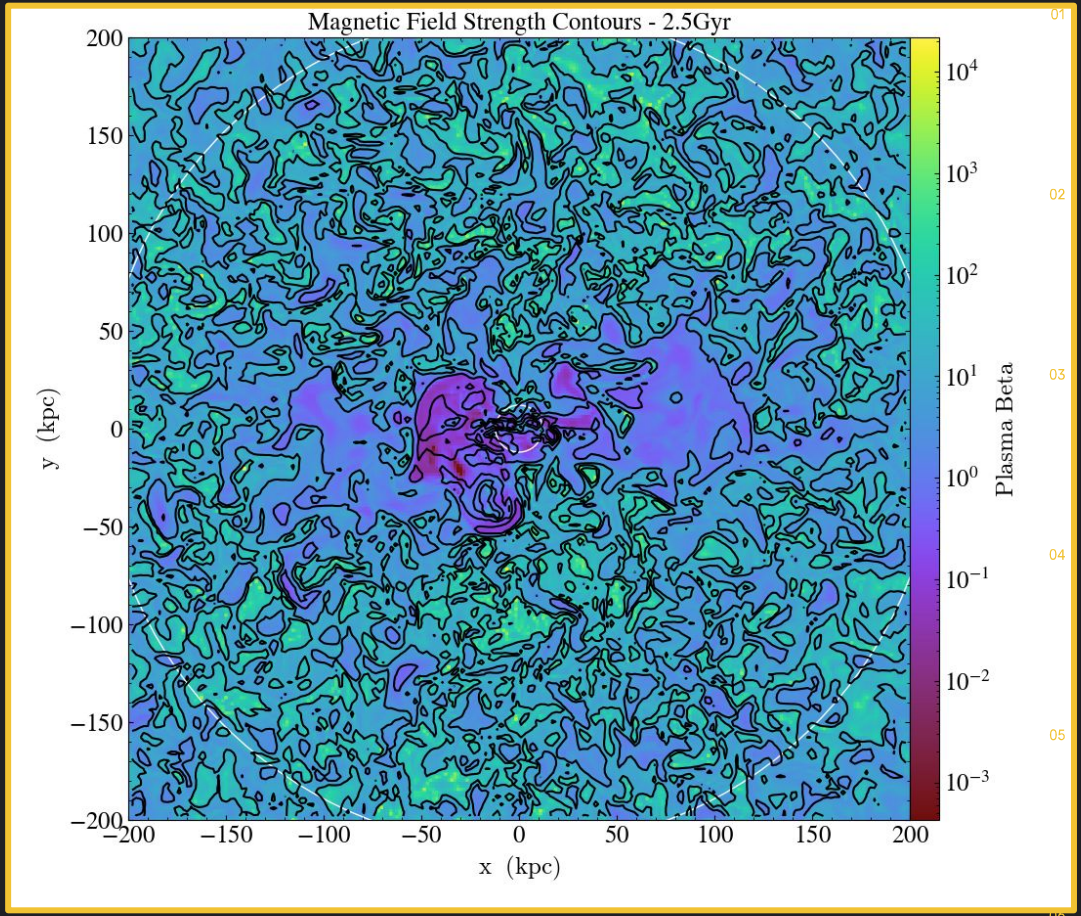




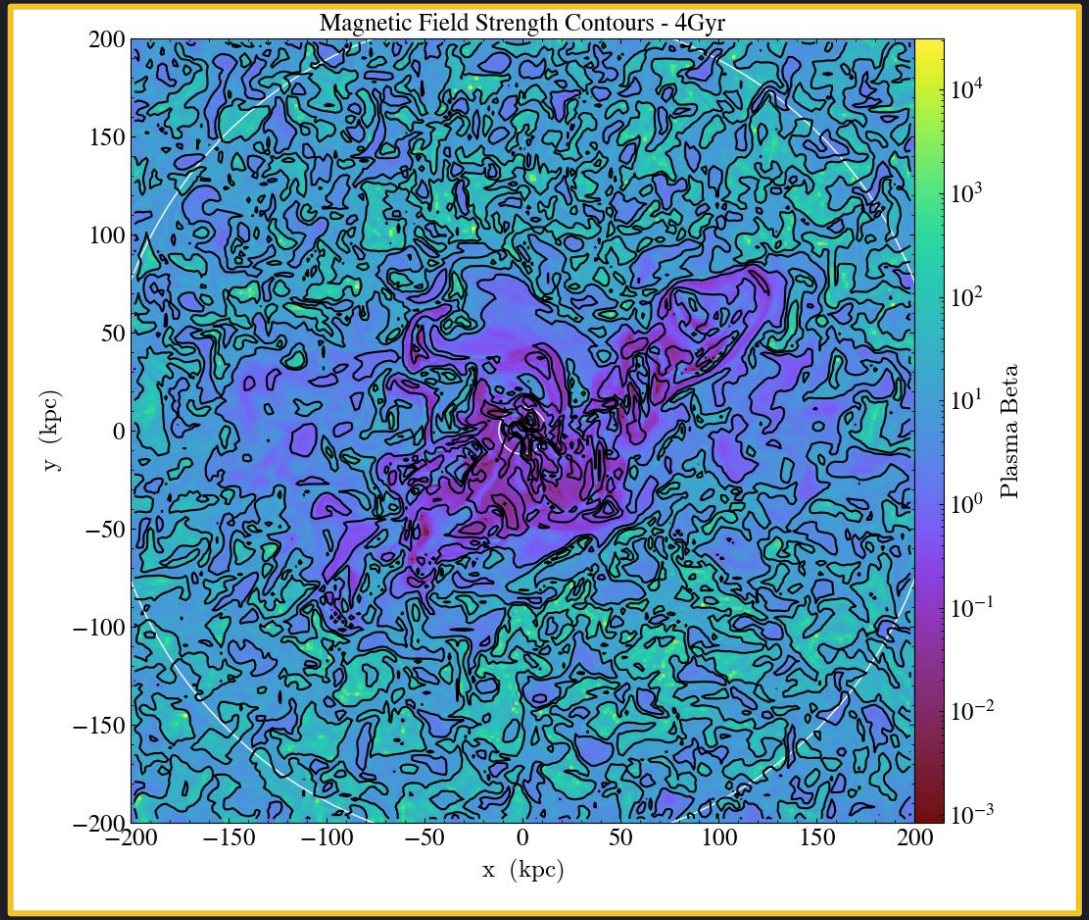
Mach-Alfven Number: A Good Test for Saturation Phenomenology?



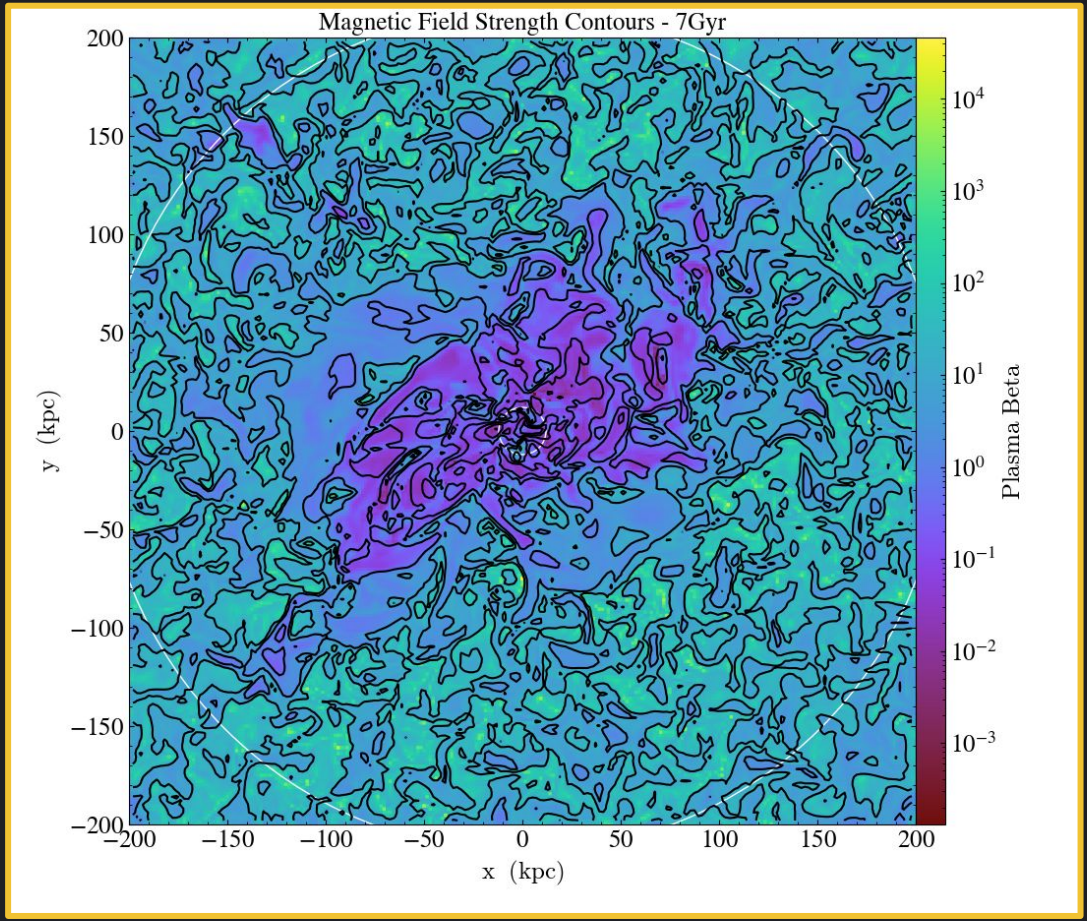
Visual Inspection



Visual Inspection



Visual Inspection



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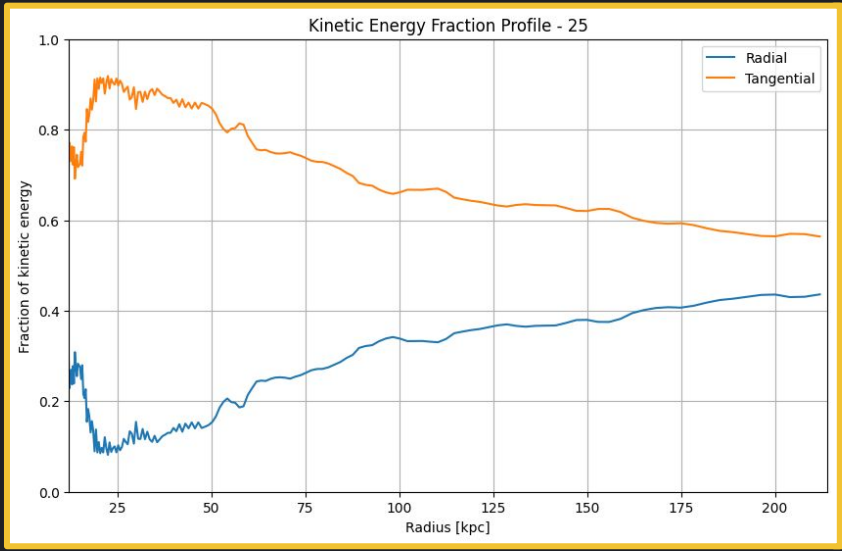
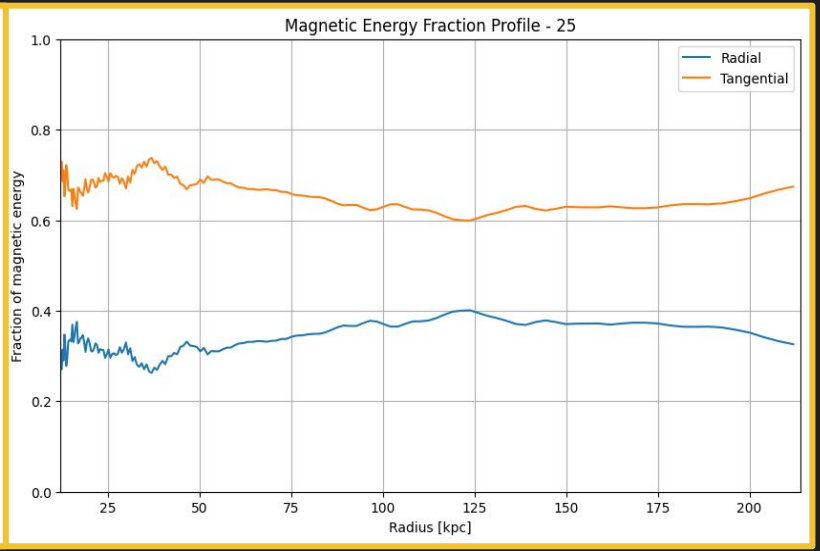
Outflows: Which Components Dominate, and how do they Affect Saturation?





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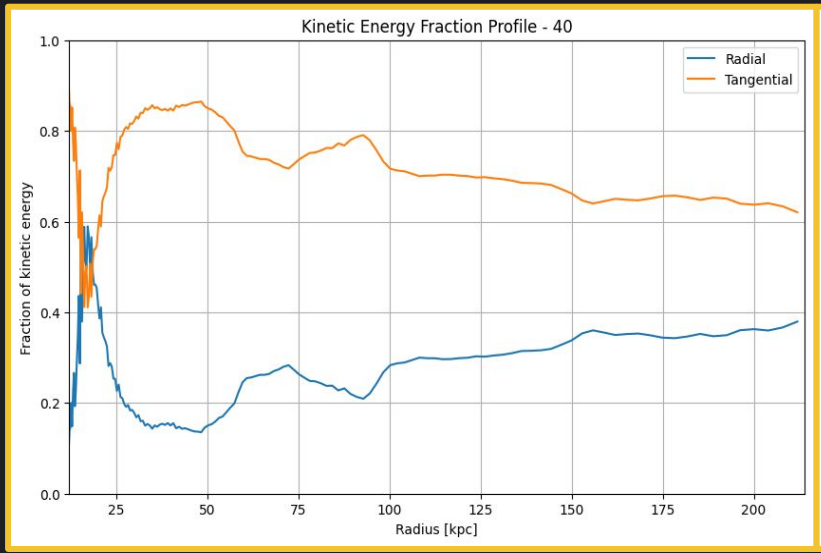

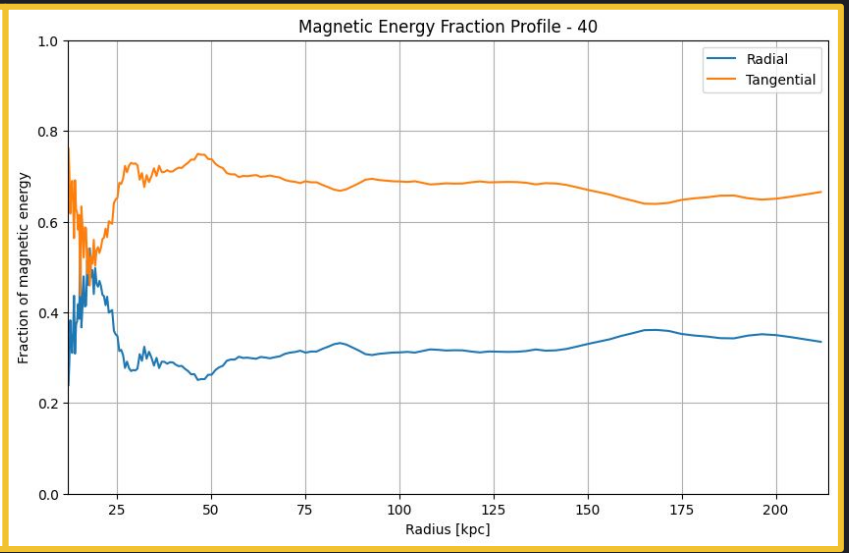

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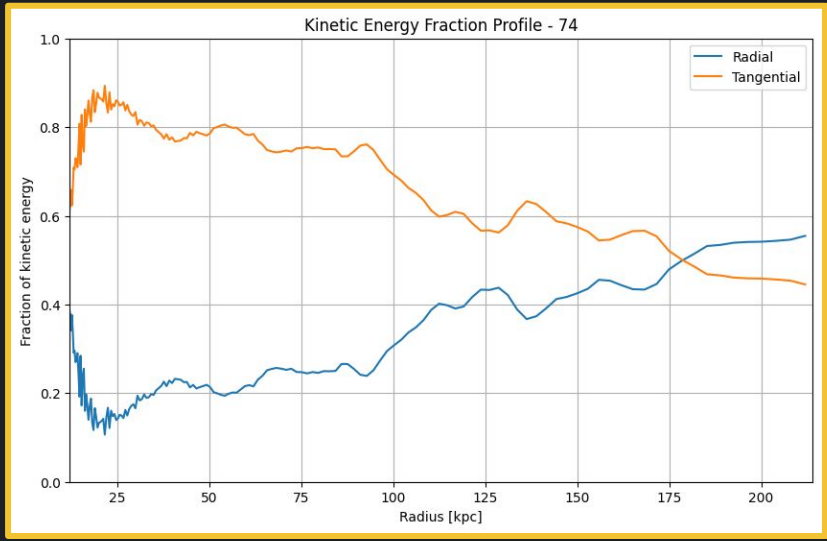
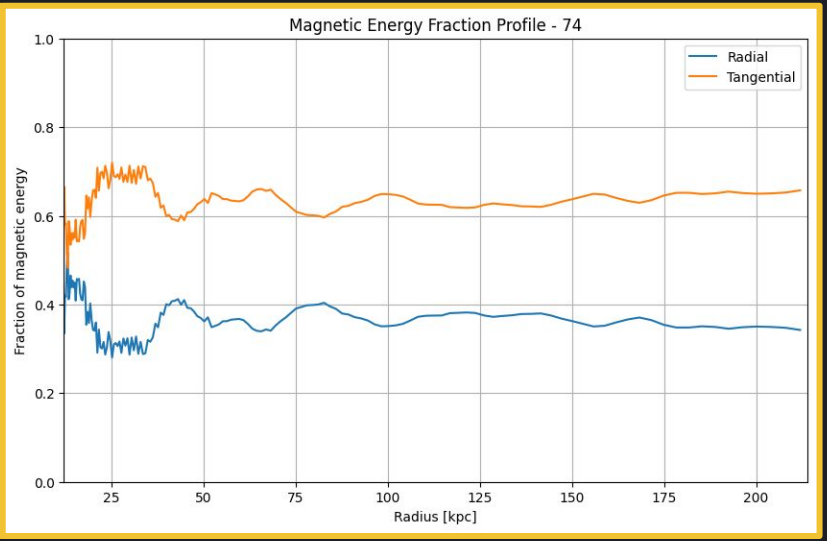
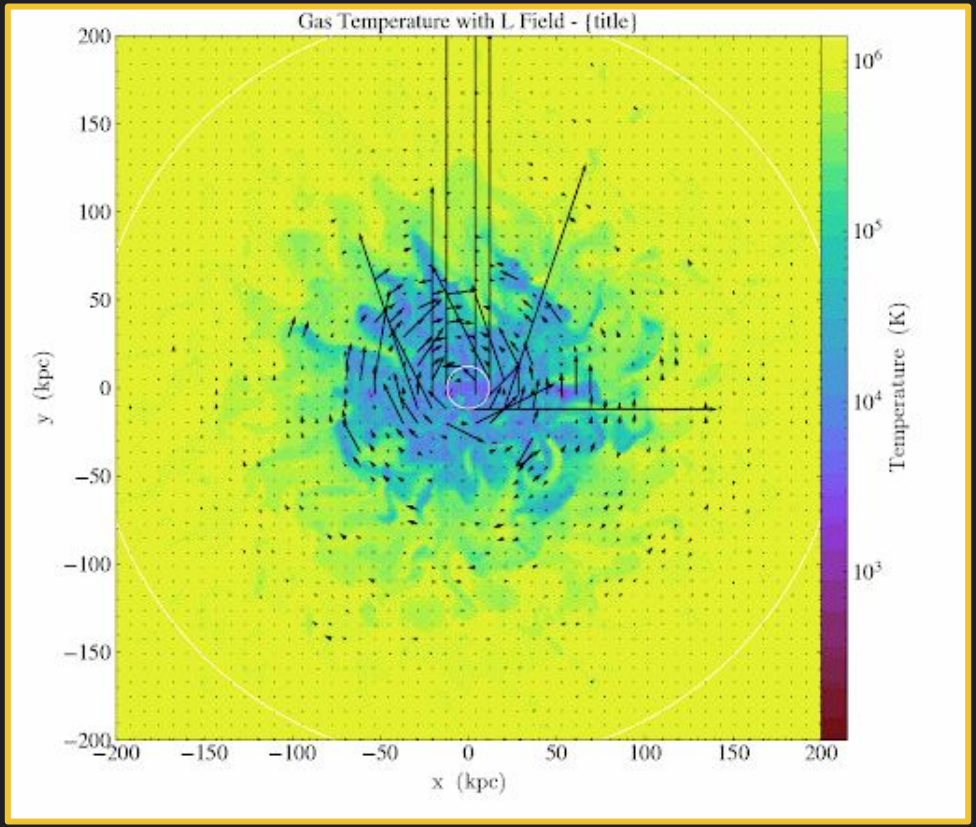
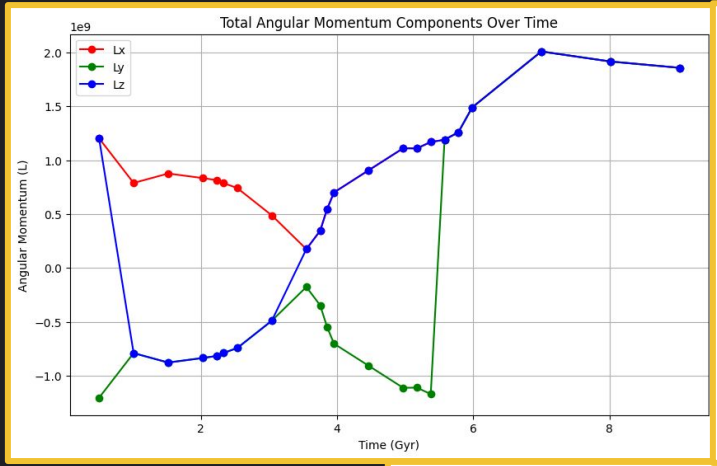
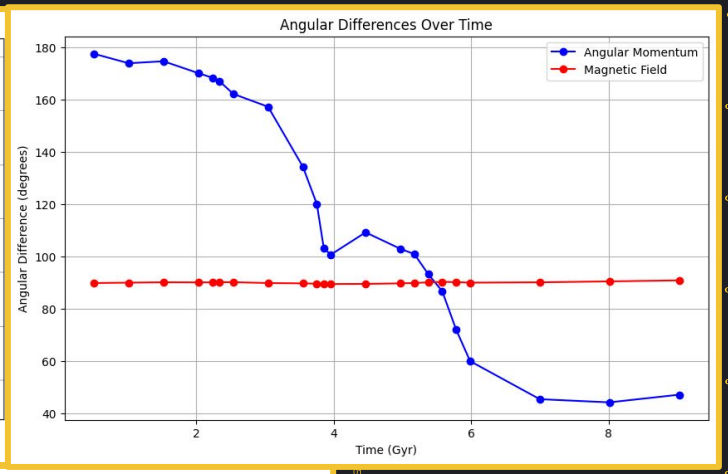


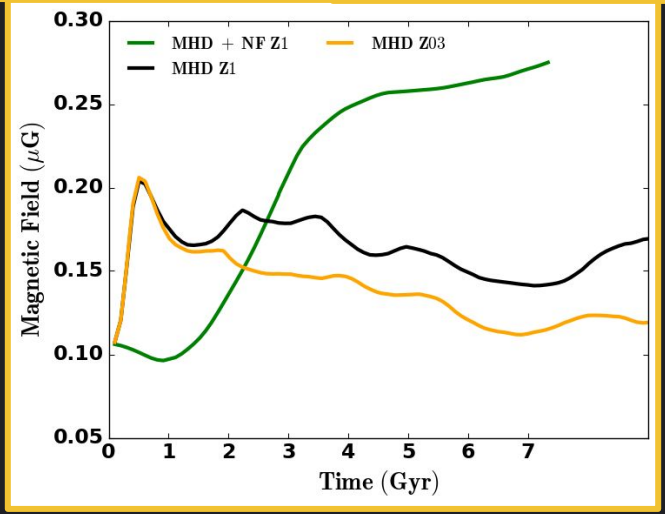


Cloud Formation and Disk Instability




Total Angular Momentum Components Over Time
1e9
Lx
Ly
Lz
Angular Momentum (L)
Time (Gyr)


Angular Differences Over Time
Angular Momentum
Magnetic Field
Angular Difference (degrees)
Time (Gyr)


MHD + NF Z1
MHD Z03
MHD Z1
Magnetic Field (μG)
Time (Gyr)

Results

Saturation Phenomenology:

1. Turbulence causes disk instability
2. Magnetic field lines remain frozen
3. Disk instability causes warping(!)
4. Ejections virialise CGM and saturate B field

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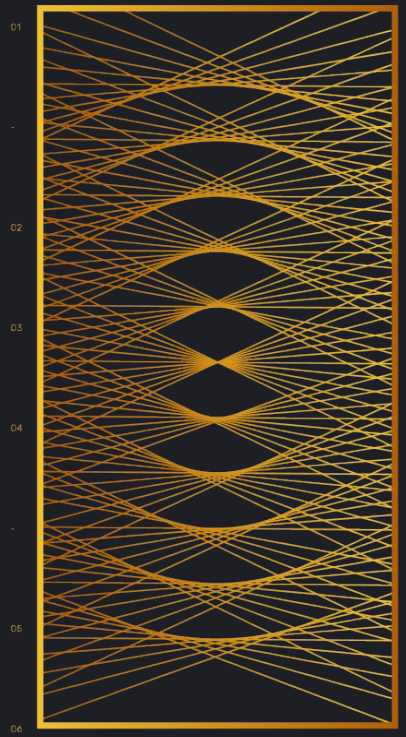

04.

Future Research

Where are we, where are we going?

Future Extensions

1. Explore parameter space: different seed B fields, metallicities, metallicity gradients
2. Extension to star and AGN particles, with simulated SNe shocks
3. Extension to large cosmological simulations: explore more MW-type galaxies with richer diversity in metallicity and morphology
4. How does the CGM of merging galaxies evolve?
5. Magnetic draping in clouds: how does this affect structure?

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Acknowledgement

- **Prof. Buie** for helpful discussions, support, and running simulations
- Texas Advanced Computing Center (**TACC**), N**ASA Pleiades** Supercomputer for computational support
- **MAIHEM**, **YT** Plotting for open-source software
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- **URSI**, **Shelby Davis Foundation** and **KNAC** for funding our research.

Questions?

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